

ESCUELA SUPERIOR POLITÉCNICA DEL LITORAL

**Faculty of Electrical and Computer Engineering
FIEC - ESPOL**

**SOFTWARE ENGINEERING II
Dream Team – Report 2nd Evaluation**

Team #2

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GUAYAQUIL - ECUADOR

Year: 2024

Software System Executive Summary

Problematic

The lack of interactive and engaging entertainment in public spaces such as shopping malls, waiting rooms, and leisure areas. Currently, most touch kiosks in these places offer limited functionalities and do not fully utilize the potential of technology to provide dynamic entertainment experiences.

Proposed solution

It is proposed to develop a Web Games Application for touch kiosks that offers a wide range of variety of interactive and entertaining games. These games will be designed to be accessible

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Figure 2: Proof after using the coding standard in the mobile application, demonstrating there are no problems.

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1. Introduction

1. Problem Description

The lack of interactive and engaging entertainment in public spaces such as shopping malls, waiting rooms, and leisure areas.

2. Problem Justification

Most touch kiosks in these places offer limited functionalities and do not fully utilize the potential of technology to provide dynamic entertainment experiences.

3. Objectives

3.1. General Objectives

Develop and deploy a web application for games designed for touch kiosks, aiming to provide an interactive and engaging experience in commercial environments. The goal is to enhance user satisfaction by rewarding winners, generating greater interest, and motivating potential customers to choose their business over other available options.

3.2. Specific Objectives

- Develop an intuitive and attractive user interface that adapts to the touch screen capabilities of the kiosk and facilitates navigation between different games.
- Design and develop a diverse collection of games covering different genres and difficulty levels, ensuring they are accessible to people of all ages.
- Collaborate with public space owners to identify and deploy kiosks equipped with the application in strategic locations with high user traffic.

2. Repository URL

Links to the repositories:

Repository of Backend: <https://github.com/RobertoEncalada/T2-Backend-Kiosco-Touch>

Repository of Frontend: <https://github.com/RobertoEncalada/T2-Frontend-Kiosco-Touch>

3. SCRUM

3.1 Role definition

Role	Member
Scrum Master	Nick Arévalo
Developer	Irving Macías
Developer	Roberto Encalada
Developer	Jefferson Eras

3.2 Backlogs

Table 1. Backlogs

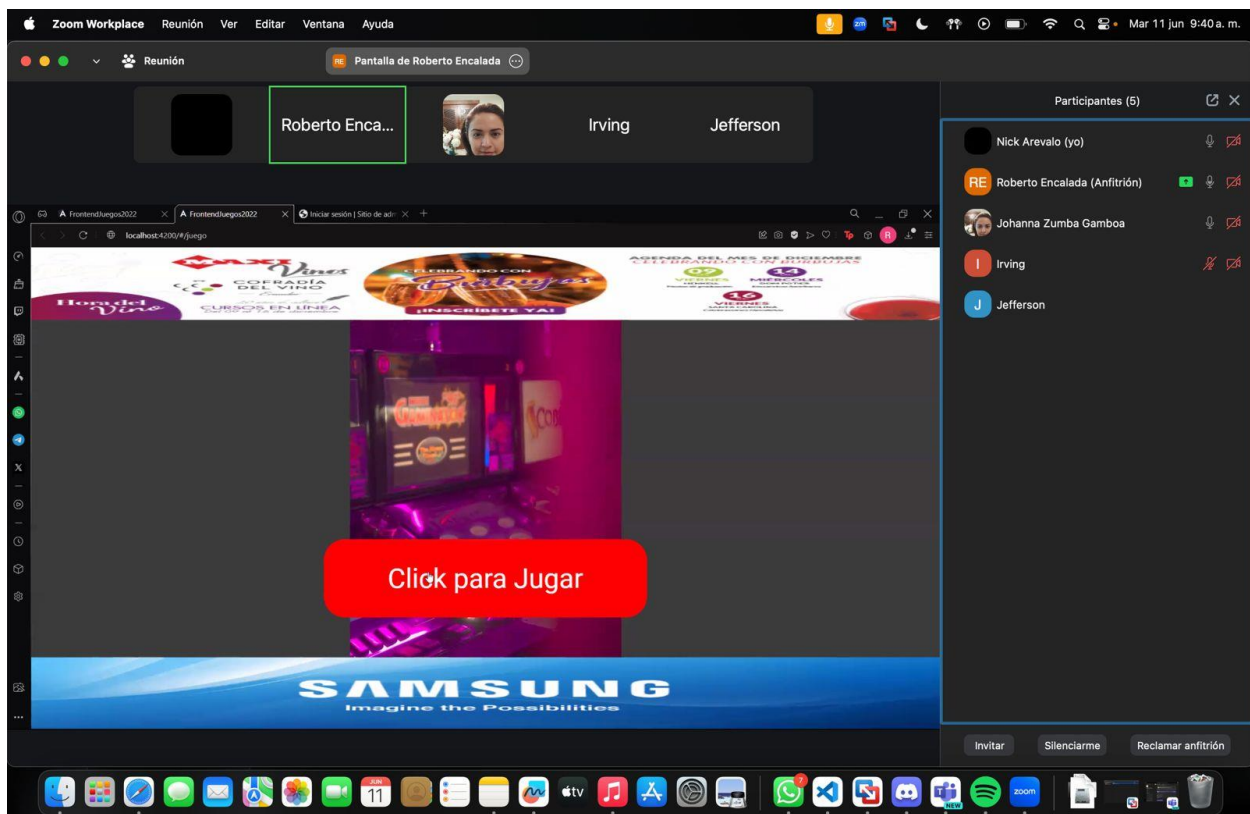
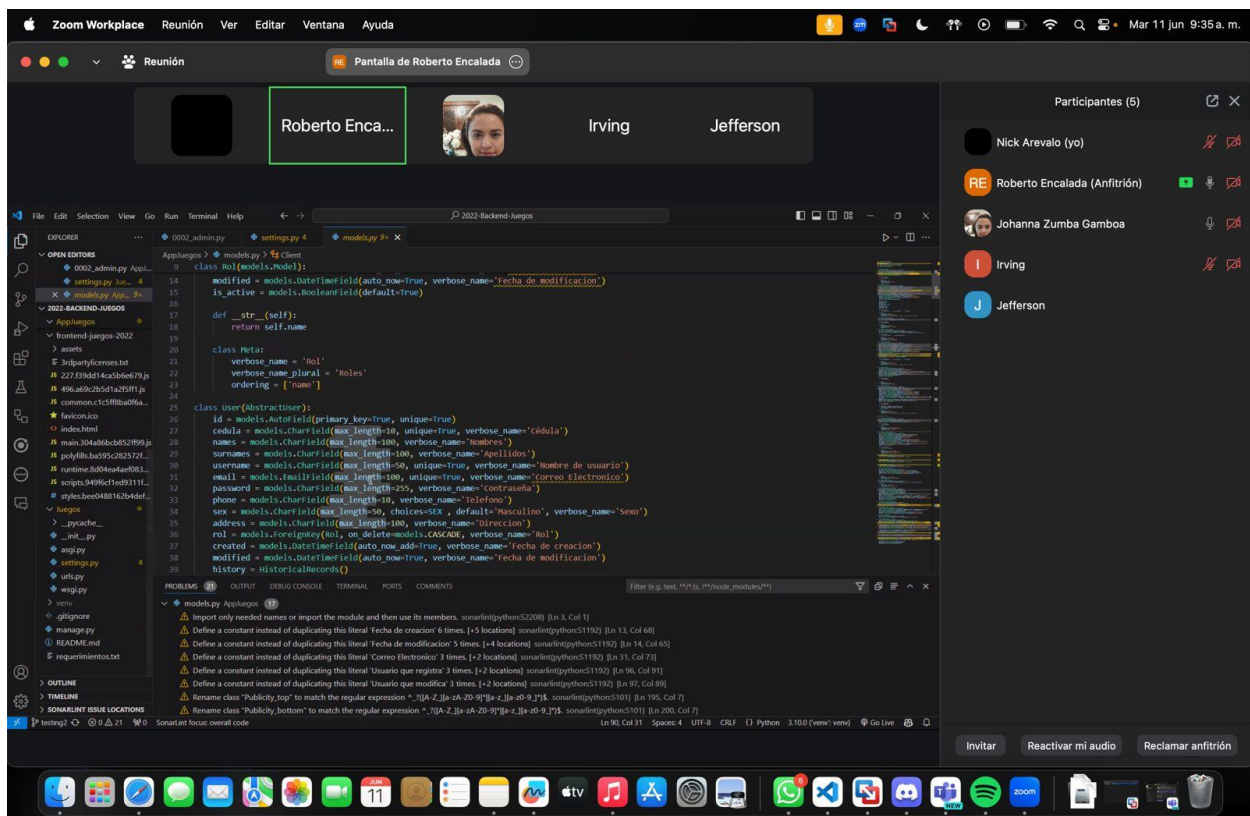
Cod	User Story for Sprint 1	Estimation (days)	Priority	Status
CL01	As a customer, I want to have a list of available games so that I can play my preferred one.	1	High	Done
CL02	As a customer, I want to play 'Roll the Dice' where I enter a number and select the "Spin" button to win a prize if it matches.	1	High	Done
CL03	As a customer, I want to play 'Guess the Door' where I select a door to win the prize inside it if it contains one.	2	Medium	Done

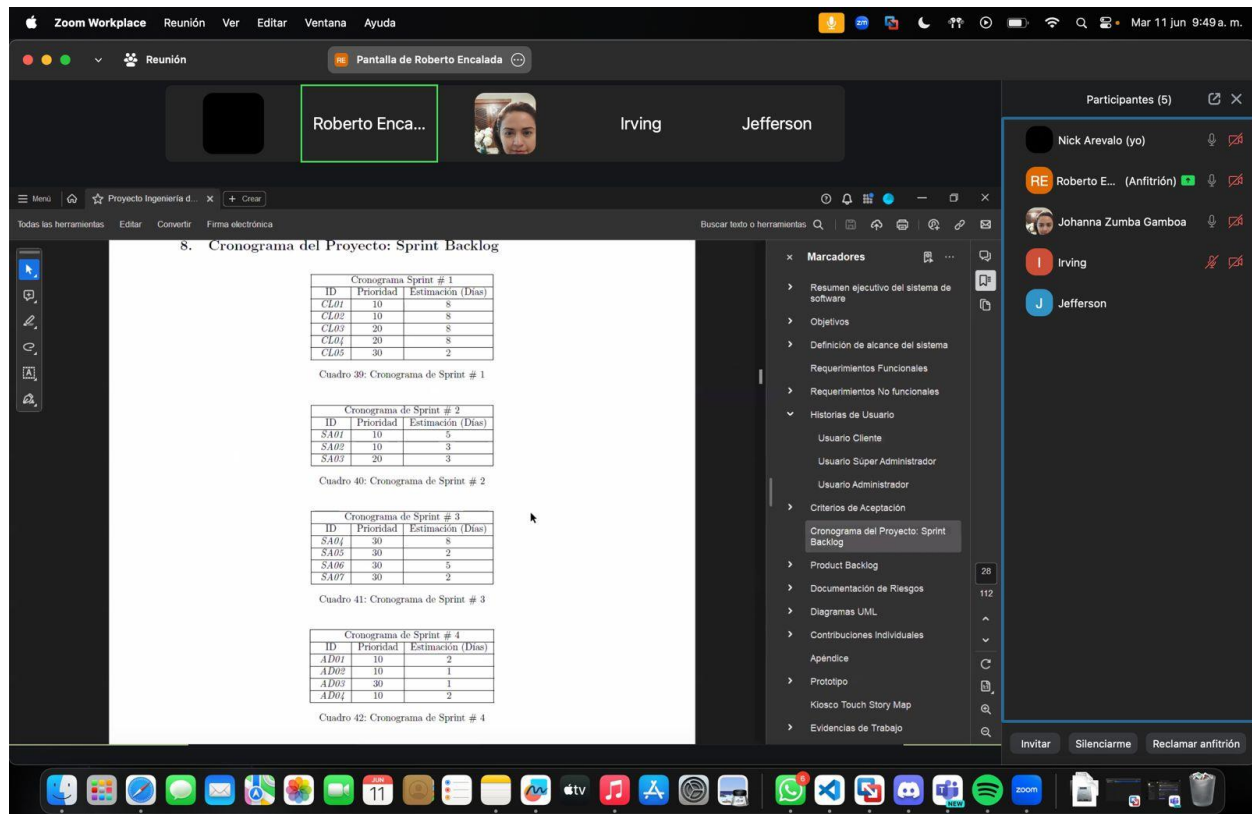
CL04	As a customer, I want to play 'Precision' where I select the "Stop" button to win a random prize if the numbers match.	3	High	Done
CL05	As a customer, I want to switch games at any time, if no game has been started previously, to try different options on the kiosk.	2	Medium	Done

3.3 Planning

Table 2. Backlogs Planning

Cod	User Story for Sprint 1	Owner	Priority	Status
CL01	As a customer, I want to have a list of available games	Roberto Encalada	5	Done
CL02	As a customer, I want to play 'Roll the Dice'	Nick Arévalo	5	Done
CL03	As a customer, I want to play 'Guess the Door'	Jefferson Eras	5	Done
CL04	As a customer, I want to play 'Precision'	Irving Macías	5	Done
CL05	As a customer, I want to switch games at any time	Nick Arévalo	5	Done





4. Online tool for teamwork management

We have chosen Asana as our online tool for teamwork management due to its robust features that enhance project coordination and communication among team members. Asana offers:

- **Task Management:** Allows us to create, assign, and track tasks efficiently, ensuring that everyone is aware of their responsibilities and deadlines.
- **Project Tracking:** Provides a visual representation of project progress through boards, timelines, and calendars, making it easy to monitor milestones and deliverables.
- **Collaboration:** Facilitates communication within the team by allowing comments, attachments, and updates directly on tasks, ensuring everyone stays informed and engaged.
- **Integration:** Easily integrates with other tools we use, such as Slack and Google Drive, streamlining our workflow and improving productivity.

We selected Asana because it helps us stay organized and ensures that we meet our project goals and deadlines.

Kiosco Touch

REIGJLNA7

Compartir

Preguntar a la IA

Personalizar

Resumen

Lista

Tablero

Cronograma

Panel

Calendario

Flujo de trabajo

Mensajes

Archivos

+

+ Agregar tarea

Filtrar

Ordenar

Agrupar por

Ocultar

...

Nombre de la tarea	Responsable	Fecha de en...	Prioridad	Estado	+
Familiarización del código	NA Nick Arevalo	26 mayo	Alto		
Independizar juego de tragamonedas	NA Nick Arevalo	29 mayo	Alto		
Agregar objetos de juego	NA Nick Arevalo	30 mayo	Medio		
Realizar pantalla de listado de juegos	NA Nick Arevalo	3 jun	Alto		
Funcionalidad juego RollTheDice	NA Nick Arevalo	6 jun	Alto		
Funcionalidad juego GuessTheDoor	NA Nick Arevalo	9 jun	Alto		
Funcionalidad juego PrecisionGame	NA Nick Arevalo	12 jun	Alto		
Realizar opción de retroceso	NA Nick Arevalo	15 jun	Bajo		
Validación de retroceso en juego	NA Nick Arevalo	Ayer	Medio		
Actualizar pantalla de listado de juegos	NA Nick Arevalo	4 jun	Medio		

5. Coding Standard Documentation and detecting and Enforcing Coding Standards

We have utilized **SonarLint** for detecting and enforcing coding standards in our project. SonarLint is an IDE extension that helps developers detect and fix quality issues as they write code. It ensures that the code adheres to predefined standards, thus improving the overall quality and maintainability of the software.

Description of the Tool:

- **SonarLint** integrates seamlessly with popular IDEs such as IntelliJ IDEA, Eclipse, and Visual Studio. It provides real-time feedback on code quality and helps developers identify and resolve issues early in the development process.

Rules Used:

- We have configured SonarLint to follow standard coding guidelines, including:
 - **Code Smells:** Identifies potential problems in the code that could lead to bugs or poor performance.
 - **Bugs:** Detects actual errors that can cause the software to behave incorrectly.
 - **Vulnerabilities:** Finds security weaknesses in the code that could be exploited by attackers.
 - **Code Coverage:** Ensures that a sufficient percentage of the code is covered by

6. Preemptive Error Detection

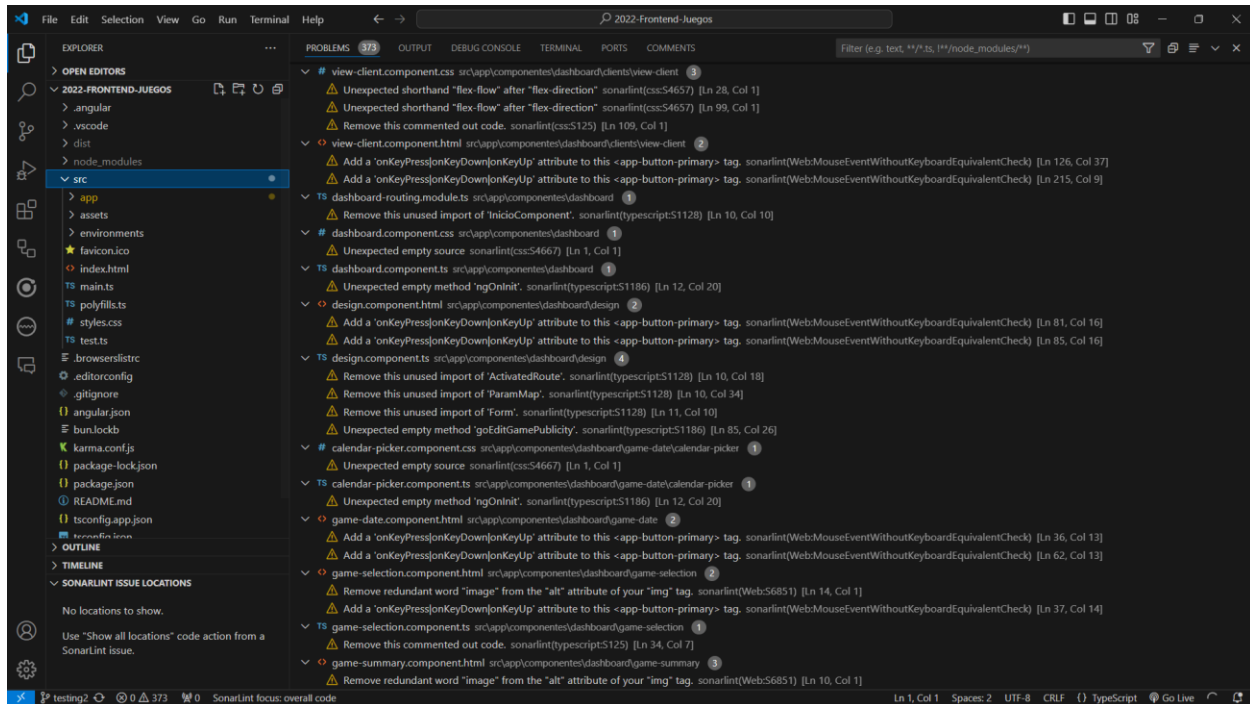


Figure 1: Proof before using the coding standard in the mobile application, demonstrating all the problems.

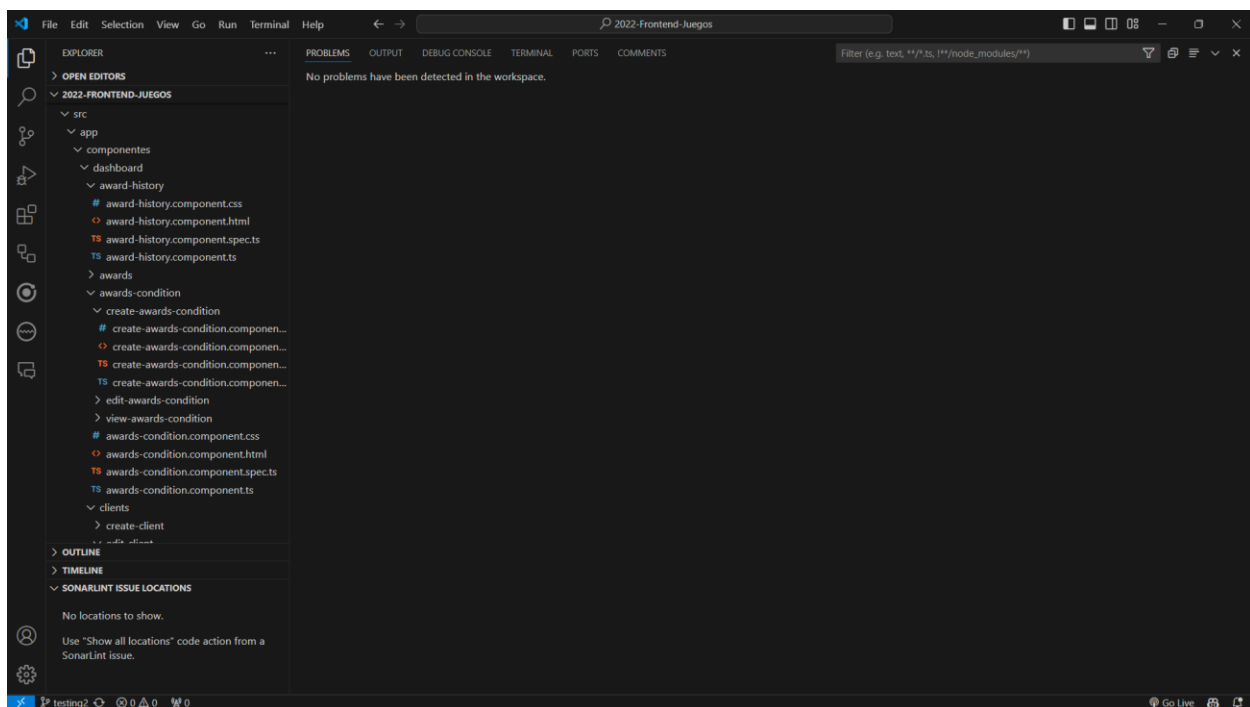


Figure 2: Proof after using the coding standard in the mobile application, demonstrating there are no problems.

7. Definition and Execution of Test Cases

The test cases have been defined and executed following the Normal Boundary Value Testing technique. Below are the details of the test cases and equivalence classes:

1. QR CODE (QR)

Test Case	Description	Example
CE1	Valid QR Code (6 numerics digits)	777888
CE2	No valid QR Code (less than 6 numerics digits)	46491
CE3	No valid QR Code (more than 6 numerics digits)	1269571
CE4	No valid QR Code (contains a nonvalid character.)	A1B2-C3D4E

2. CURRENT DATE (DATE)

Format: YYYY-MM-DD, within the year 2024.

Test Case	Description	Example
CE1	Valid date (within the range 2024-01-01 a 2024-12-31)	2024-06-15
CE2	Invalid date (before 2024-01-01)	2023-12-31
CE3	Invalid date (after 2024-12-31)	2025-01-01
CE4	Invalid date (incorrect format)	2024-06-15

3. CURRENT HOUR (HOUR)

- *Format:* HH in 24 hours.

Test Case	Description	Example
CE1	Valid Hour (In the range 00:00 a 23:59)	12:00
CE2	Invalid Hour (Incorrect format)	12:60
CE3	Invalid Hour (Out the range 00:00 a 23:59)	24:00
CE4	Invalid Hour (contains non numeric characters)	12:AB

Test Case Results

Test Case 1: Validation of the QR Code

Test case ID:	TC-QR-01
Test Priority:	1
Module Name:	QR Module
Test title:	QR Minimum
Test Description:	Allow the customer to validate the QR code before playing
Pre-condition:	The customer must have obtained a ticket and have a valid QR code
Expected output:	QR code validated successfully
Actual output:	QR code validated successfully
Status:	Pass

Screenshot: QR Code Validation (Before Test)



Screenshot: QR Code Validation (After Test)



Test Case 2: Prize determination

Test case ID:	CP-2
Test Priority:	2
Module Name:	Game
Test title:	Prize Determination
Test Description:	Determines if the customer wins a prize based on date, time and probability
Pre-condition:	The customer has validated their QR code and is within the game date and time
Expected output:	Correct prize determination according to established conditions
Actual output:	Correct prize determination according to established conditions
Status:	Pass

Test Case 3: Prize verification

Test case ID:	CP-3
Test Priority:	3
Module Name:	Prizes
Test title:	Prize Verification
Test Description:	Allows the customer to verify if they have won a prize
Pre-condition:	The customer has validated their QR code and is within the game date and time
Expected output:	Message indicating whether you have won a prize or not
Actual output:	Message indicating whether you have won a prize or not
Status:	Pass

Screenshot: Prize Determination (Before the Test)



Screenshot: Prize Determination (After the Test)

Winner



Losser



Decision Table for Project

Conditioned Award	T	T	T	T	F	F	F	F
Random Number [1 - 1000] 1 <= n <= 200	T	T	F	F	T	T	F	F
Win Award	T	F	T	F	T	F	T	F
Loser Result		X	X	X	X	X	X	X
Winner Result	X							

Number of Test cases: $2^n = 2^3 = 8$

Redundancy detection

<i>Conditioned Award</i>	<i>T</i>	<i>T</i>	<i>T</i>	<i>T</i>	<i>F</i>	<i>F</i>	<i>F</i>	<i>F</i>
<i>Random Number [1 - 1000]</i> <i>1 <= n <= 200</i>	<i>T</i>	<i>T</i>	<i>F</i>	<i>F</i>	<i>T</i>	<i>T</i>	<i>F</i>	<i>F</i>
<i>Win Award</i>	<i>T</i>	<i>F</i>	<i>T</i>	<i>F</i>	<i>T</i>	<i>F</i>	<i>T</i>	<i>F</i>
<i>Loser Result</i>		<i>X</i>	<i>X</i>	<i>X</i>	<i>X</i>	<i>X</i>	<i>X</i>	<i>X</i>
<i>Winner Result</i>	<i>X</i>							

Number of Test cases: $2^n = 2^3 = 8$

Can be reduced to 2

Decision Table after Analysis

<i>Prize Type</i>	<i>Conditioned Award</i>		<i>Don't Conditioned Award</i>	
<i>Random Number</i>	<i>[1-200]</i>	<i>[201 - 1000]</i>	<i>[1-200]</i>	<i>[201 - 1000]</i>
<i>Prize Exists</i>	<i>Exists</i>	<i>Don't Exists</i>	<i>Exists</i>	<i>Don't Exists</i>
<i>Loser Result</i>		<i>X</i>		
<i>Winner Result</i>	<i>X</i>			

7.1 Unit Testing Implementation

To ensure code quality, unit testing was implemented for each component, and more extensive testing was implemented for critical system components, focusing on input validation and key business logic, such as QR code validation, date and time verification. and determination of prizes, for this we use the following tools:

- Tools Used

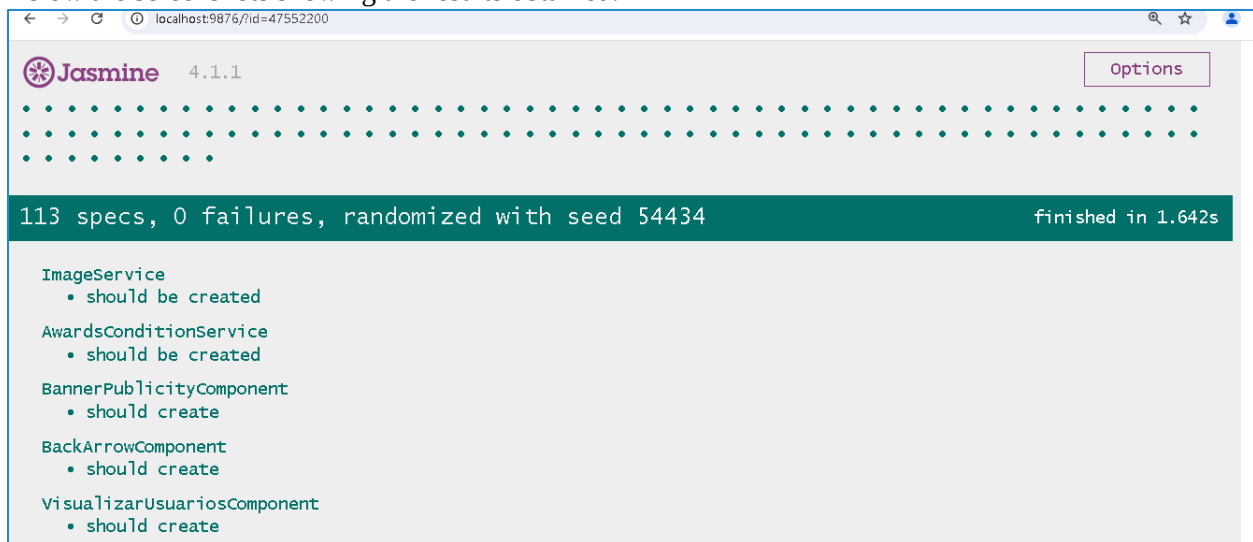
Jasmine: Framework for writing unit tests that allows you to define test cases and expectations in a structured and readable way.

Karma: Test runner that runs tests in different browsers, ensuring that the code is compatible with various environments.

Angular CLI: Used to run ng test, integrating Jasmine and Karma into the unit testing process.

- Test Results

Unit tests were successfully executed using ng test, and detailed reports of the results were generated. Below are screenshots showing the results obtained:



1. Extensive Test Results for "ScanViewComponent"

- QR Code Validation Test Cases

The test cases defined for QR code validation (CE1 to CE4) have been executed and verified in the ScanViewComponent component. Test results confirmed that the component correctly handles valid and invalid codes and provides appropriate feedback to users.

```
// Test Cases

it('should continue to game with valid QR code (CE1)', async () => {
  mockKeyControllerService.getCode.and.returnValue('777888'); // Valid QR Code
  mockGameLogicService.verifyTicket.and.returnValue(Promise.resolve(true));
  await component.continueToGame();
  expect(mockRouter.navigate).toHaveBeenCalled();
  expect(mockGameLogicService.playGame).toHaveBeenCalled();
  expect(sessionStorage.getItem('selection_game')).toBe('selection_game');
});

it('should show error for QR code with less than 6 digits (CE2)', async () => {
  mockKeyControllerService.getCode.and.returnValue('46491'); // Invalid QR Code (less than 6 digits)
  mockGameLogicService.verifyTicket.and.returnValue(Promise.resolve(false));

  await component.continueToGame();
  expect(mockConfirmDialogService.error).toHaveBeenCalled();
  expect(mockKeyControllerService.clearCode).toHaveBeenCalled();
});

it('should show error for QR code with more than 6 digits (CE3)', async () => {
  mockKeyControllerService.getCode.and.returnValue('1269571'); // Invalid QR Code (more than 6 digits)
  mockGameLogicService.verifyTicket.and.returnValue(Promise.resolve(false));
  await component.continueToGame();
  expect(mockConfirmDialogService.error).toHaveBeenCalled();
  expect(mockKeyControllerService.clearCode).toHaveBeenCalled();
});

it('should show error for QR code with non-numeric characters (CE4)', async () => {
  mockKeyControllerService.getCode.and.returnValue('A1B2-C3D4E'); // Invalid QR Code (non-numeric characters)
  mockGameLogicService.verifyTicket.and.returnValue(Promise.resolve(false));
  await component.continueToGame();
  expect(mockConfirmDialogService.error).toHaveBeenCalled();
  expect(mockKeyControllerService.clearCode).toHaveBeenCalled();
});
```

6 specs, 0 failures, randomized with seed 18457

ScanViewComponent

- should change view to input code state
- should show error for QR code with more than 6 digits (CE3)
- should show error for QR code with non-numeric characters (CE4)
- should show error for QR code with less than 6 digits (CE2)
- should create scan view component
- should continue to game with valid QR code (CE1)

2. Extensive Test Results for "CreateGameDateComponent"

- Date and Time Validation Test Cases

The test cases defined for date and time validation (CE1 to CE4) have been executed and verified in CreateGameDateComponent. Test results confirmed that the component correctly handles valid and invalid date and time ranges, ensuring that the selected start and end times are within the allowed period.

```

it('should calculate startDate and endDate correctly when times are within the same day', async () => {
  component.beginDate = new Date(2023, 7, 16); // 16 de agosto de 2023
  component.finishDate = new Date(2023, 7, 16); // 16 de agosto de 2023

  component.form.patchValue({
    startTime: { hour: 10, minute: 30 }, // 10:30 AM
    endTime: { hour: 15, minute: 45 } // 3:45 PM
  });

  await component.changetime();

  expect(component.startDate).toEqual(new Date(2023, 7, 16, 10, 30)); // 16 de agosto de 2023, 10:30 AM
  expect(component.endDate).toEqual(new Date(2023, 7, 16, 15, 45)); // 16 de agosto de 2023, 3:45 PM
});

it('should calculate startDate and endDate correctly when start and end times are on different days', async () => {
  component.beginDate = new Date(2023, 7, 16); // 16 de agosto de 2023
  component.finishDate = new Date(2023, 7, 17); // 17 de agosto de 2023

  component.form.patchValue({
    startTime: { hour: 22, minute: 15 }, // 10:15 PM
    endTime: { hour: 1, minute: 30 } // 1:30 AM (next day)
  });

  await component.changetime();

  expect(component.startDate).toEqual(new Date(2023, 7, 16, 22, 15)); // 16 de agosto de 2023, 10:15 PM
  expect(component.endDate).toEqual(new Date(2023, 7, 17, 1, 30)); // 17 de agosto de 2023, 1:30 AM
});

it('should calculate startDate and endDate correctly when end time is earlier on the same day', async () => {
  component.beginDate = new Date(2023, 7, 16); // 16 de agosto de 2023
  component.finishDate = new Date(2023, 7, 16); // 16 de agosto de 2023

  component.form.patchValue({
    startTime: { hour: 14, minute: 0 }, // 2:00 PM
    endTime: { hour: 11, minute: 30 } // 11:30 AM (earlier in the same day)
  });

  await component.changetime();
  expect(component.startDate).toEqual(new Date(2023, 7, 16, 14, 0)); // 16 de agosto de 2023, 2:00 PM
  expect(component.endDate).toEqual(new Date(2023, 7, 16, 11, 30)); // 16 de agosto de 2023, 11:30 AM
});

```

As for the validations of the date and time format, it has already been previously validated by the MatCalendar and NgbTimepicker components.

4 specs, 0 failures, randomized with seed 55571

CreateGameDateComponent

- should create

changetime

- should calculate startDate and endDate correctly when times are within the same day
- should calculate startDate and endDate correctly when end time is earlier on the same day
- should calculate startDate and endDate correctly when start and end times are on different days

3. Extensive Test Results for "GameLogicService"

GameLogicService is the main component responsible for determining whether a player is eligible to receive a prize based on predefined conditions.

The logic involves multiple factors:

1. Probabilistic calculations.
2. Date range validation.
3. Prize stock.

Using mocks, we are going to simulate the services that GameLogicService uses to avoid real calls to our BACKEND and control the behavior of the dependencies.

```
describe('getPrize', () => {
  it('should return an award from the "Común" category if rd_number <= 800', async () => {
    service['winProb'] = 1.0; // 100% de probabilidad de ganar
    spyOn(Math, 'random').and.returnValue(0.5);
    const result = await service.getPrize();
    expect(result).toBeTruthy();
    if (result) {
      expect(result.length).toBeGreaterThan(0);
      expect(result[0].category).toBe('Común');
    }
  });

  it('should return an award from the "Rara" category if rd_number > 800 and <= 950', async () => {
    service['winProb'] = 1.0; // 100% de probabilidad de ganar
    spyOn(Math, 'random').and.returnValue(0.85);
    const result = await service.getPrize();

    expect(result).toBeTruthy();
    if (result) {
      expect(result.length).toBeGreaterThan(0);
      expect(result[0].category).toBe('Rara');
    }
  });

  it('should return an award from the "Épica" category if rd_number > 950 and <= 990', async () => {
    service['winProb'] = 1.0; // 100% de probabilidad de ganar
    spyOn(Math, 'random').and.returnValue(0.975);

    const result = await service.getPrize();

    expect(result).toBeTruthy();
    if (result) {
      expect(result.length).toBeGreaterThan(0);
      expect(result[0].category).toBe('Épica');
    }
  });
});
```



```
it('should return an award from the "Legendaria" category if rd_number > 990 and <= 1000', async () => {
  service['winProb'] = 1.0; // 100% de probabilidad de ganar
  spyOn(Math, 'random').and.returnValue(0.995);
  const result = await service.getPrize();

  expect(result).toBeTruthy();
  if (result) {
    expect(result.length).toBeGreaterThan(0);
    expect(result[0].category).toBe('Legendaria');
  }
});
```

```
it('should return null if no awards are available in the category', async () => {
  service['winProb'] = 1.0; // 100% de probabilidad de ganar
  spyOn(Math, 'random').and.returnValue(0.5);

  // Simula que no hay premios en la categoría
  awardsService.getFilterAward.and.callFake((filter: string) => of([]));
  const result = await service.getPrize();
  expect(result).toBeNull(); // Verifica que el resultado sea null
});
```

```
it('should return null if the random number falls outside the winning probability', async () => {
  service['winProb'] = 0;
  spyOn(Math, 'random').and.returnValue(0);
  const result = null;
  expect(result).toBeNull(); // Verifica que el resultado sea null
});
```

6 specs, 0 failures, randomized with seed 19268

GameLogicService

getPrize

- should return an award from the "Común" category if rd_number <= 800
- should return null if no awards are available in the category
- should return an award from the "Legendaria" category if rd_number > 990 and <= 1000
- should return an award from the "Rara" category if rd_number > 800 and <= 950
- should return an award from the "Épica" category if rd_number > 950 and <= 990
- should return null if the random number falls outside the winning probability

7.2 Risk-Based Testing Details:

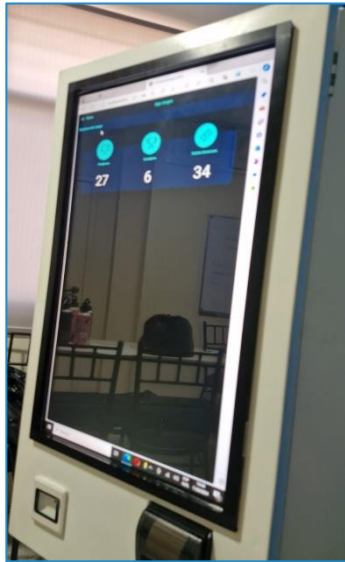
In the project, several risks are identified that could affect the quality of the final product:

7.2.1 Risk-Based Testing Content:

- **Risk Identification:**

Technical Risks:

1. **Integration with Backend:** There is a risk that the integration between the Angular frontend and the Django backend may face issues such as mismatched data formats, incorrect API responses, or communication failures.
2. **Hardware Integration:** Risks associated with integrating hardware components like the 32-inch TV, label printer, and infrared QR reader with the software.



Functional Risks:

In the development of our web application for the touch kiosk, it is essential to identify and manage functional risks that may affect operation and user experience. The system is divided into two key modules: Game Modules and Admin Dashboard, the risks are detailed below

Functional Risks Table

Aspect	Description	Potential Impact
Game Mode Selection	Issues with the logic for selecting the correct game mode.	Affected user experience, potential errors in the game.
Loading Selected Game	The selected game does not load correctly.	Interruption of user experience, game unavailability.
Allowed Attempts Adjustment	Errors in adjusting the number of allowed attempts.	Affects game difficulty and user experience.
Win Probability Adjustment	Issues with setting the probability of winning.	Affected user experience, possible gameplay imbalances.
Prize Loading	The system does not load prizes correctly.	Errors in prize delivery, user dissatisfaction.

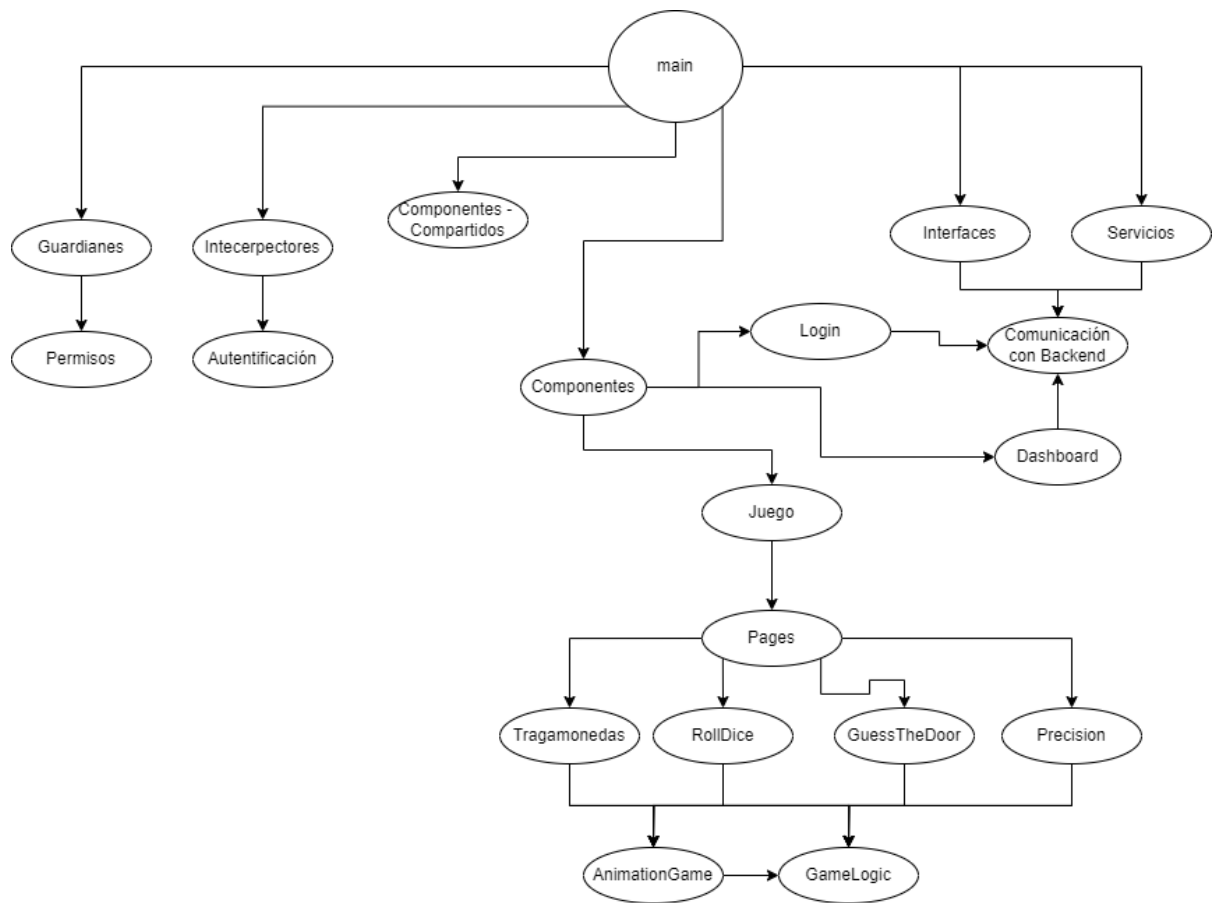
Manage Games	Problems with game management.	Difficulties in updating or removing games.
Roles	Issues with assigning and managing user roles.	Unauthorized access or permission management problems.

Non-Functional Risks Table

Aspect	Description	Objective
Load Testing	Evaluate system performance under a representative load.	Ensure the system handles the expected number of users.
Stress Testing	Assess how the system handles extreme load conditions.	Determine the system's breaking point.
Performance Testing	Measure response time and transactions per second.	Ensure the system meets performance requirements.
Volume Testing	Evaluate the system's handling of large volumes of data.	Confirm that the system can manage large amounts of data.
Security Testing	Assess protection against unauthorized access and malicious damage.	Ensure the system is secure against vulnerabilities.
Usability Testing	Measure user interaction efficiency and ease of use.	Ensure the system is user-friendly.
Storage Testing	Verify that the system meets specified storage objectives.	Confirm the system correctly stores data.

Connectivity Risks

- 1. Internet Loss:** The application must be tested to ensure that critical functionalities, such as game operation and printing, continue to work even if internet connectivity is lost.
- 2. Data Synchronization:** Risks associated with the synchronization of data between the kiosk and any remote servers or databases when connectivity is unstable or lost.

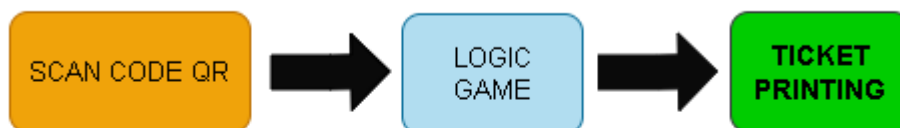


- **Test Case Prioritization:**

In the following figure, the Call Graph of our system is showed, and we can observe that the nodes “Comunicación con Backend”, “AnimationGame” and “GameLogic” are the nodes with the most quantity of indegree

The interfaces where the greatest probability of risk is focused are:

GAME MODULE



DASHBOARD MODULE



- **Mitigation Strategies:**

To ensure the robustness and reliability of the system, we have identified several key risks and formulated mitigation strategies:

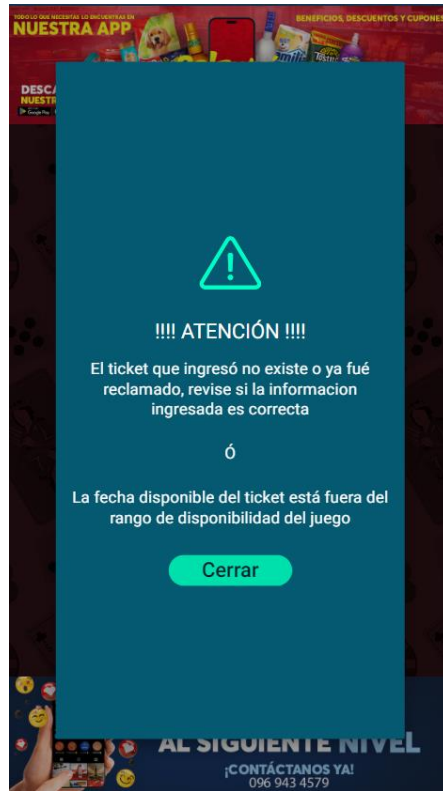
Usability Testing

To mitigate the risk of QR code scanning failure, a touch keyboard was implemented as a secondary option for manual code entry.



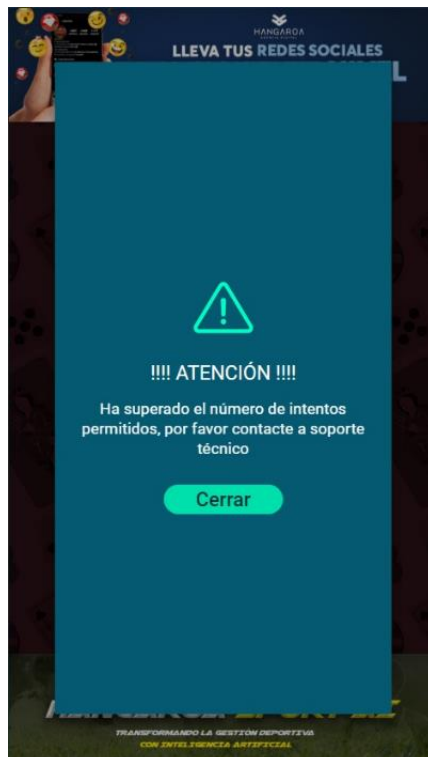
Performance Testing

To address communication issues between the frontend and backend, the game's functionality was isolated, ensuring verification is done via the ticket to load the complete functionality.

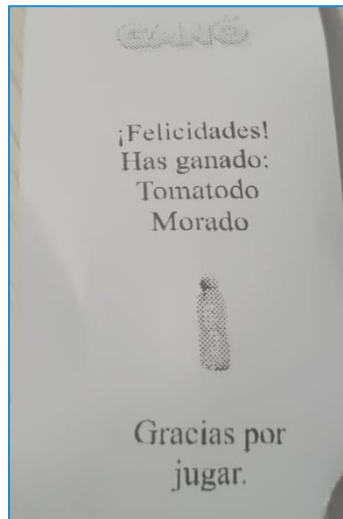


Security Testing

To prevent misuse of tickets and ensure that they cannot be used more times than allowed, we have implemented a robust control mechanism to control allowed attempts.



To address potential issues related to ticket validation and usage, the system is designed to ensure safe and accurate handling of each ticket. When a player wins, the system will physically print a ticket with a unique and valid code on the same day.



2. Risk-Based Testing Order:

Aspect	Description	Potential Impact	Priority
Game Mode Selection	Issues with the logic for selecting the correct game mode.	Affected user experience, potential errors in the game.	High
Loading Selected Game	The selected game does not load correctly.	Interruption of user experience, game unavailability.	High
Allowed Attempts Adjustment	Errors in adjusting the number of allowed attempts.	Affects game difficulty and user experience.	Medium
Win Probability Adjustment	Issues with setting the probability of winning.	Affected user experience, possible gameplay imbalances.	Medium
Prize Loading	The system does not load prizes correctly.	Errors in prize delivery, user dissatisfaction.	High
Manage Games	Problems with game management.	Difficulties in updating or removing games.	Medium
Roles	Issues with assigning and managing user roles.	Unauthorized access or permission management problems.	Low

Test Sequence

1. **High Priority:**
 - **Game Mode Selection:** Verify the correct selection of the game mode.
 - **Loading Selected Game:** Confirm that the games load correctly.
 - **Prize Loading:** Ensure that the prizes load without errors.
2. **Medium Priority:**
 - **Allowed Attempts Adjustment:** Test the correct configuration of the allowed attempts.
 - **Win Probability Adjustment:** Verify the accuracy in adjusting the probability of winning.
 - **Manage Games:** Ensure that the game management is correct and functional.
3. **Low Priority:**
 - **Roles:** Validate the correct assignment and management of user roles.
4. **Resource Allocation**

Priority	Assigned Resources	Estimated Time
High	Development Team, Senior QA	40 hours
Medium	QA Team, Junior Developer	30 hours
Low	Documentation Team, Usability Tester	

5. Results Explained or Corrected

Explanation of Results: The load testing results showed that the system could handle the expected number of users without significant issues. However, some minor issues in the user interface were identified during usability testing, which were corrected before the final implementation.

8. System Deployment

Deployment Process

The application deployment process is carried out using PythonAnywhere, a cloud hosting platform that makes it easy to deploy Python-based web applications.

Python Anywhere

Environment Configuration:

Updating code in PythonAnywhere: For this process we used our GitHub repository and proceeded to upload the code

```
 Bash console in virtualenv venv  
(venv) 23:49 ~/2022-Backend-Juegos/frontend-juegos-2022 (produccion)$ git pull  
Username for 'https://github.com': JeffErasLindao  
Password for 'https://JeffErasLindao@github.com':  
remote: Enumerating objects: 51, done.  
remote: Counting objects: 100% (48/48), done.  
remote: Compressing objects: 100% (15/15), done.  
remote: Total 29 (delta 22), reused 21 (delta 14), pack-reused 0 (from 0)  
Unpacking objects: 100% (29/29), 3.42 KiB | 6.00 KiB/s, done.  
From https://github.com/RobertoEncalada/2022-Backend-Juegos  
6fd2687..6ff4d30 local -> origin/local  
Already up to date.  
(venv) 23:52 ~/2022-Backend-Juegos/frontend-juegos-2022 (produccion)$
```

Finally we proceded with Configuration of databases:

 **pythonanywhere**
by ANACONDA

Dashboard Consoles

MySQL

Postgres

MySQL settings

Connecting:

Use these settings in your web applications.

Database host address: kioskotouch.mysql.pythonanywhere-services.com

Username: kioskotouch

URL's: Production Deployment - GAME

<https://kioskotouch.pythonanywhere.com/#!/juego>

 **pythonanywhere**
by ANACONDA

Dashboard Consoles Files Web

kioskotouch.pythonanywhere.com

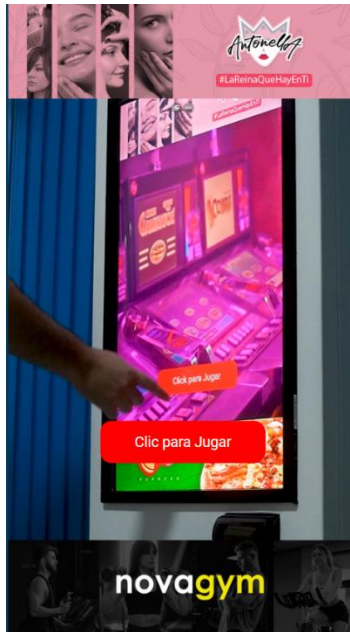
Add a new web app

Configuration for kioskotouch.pythonanywhere.com

Reload:

Reload kioskotouch.pythonanywhere.com

Best before date:

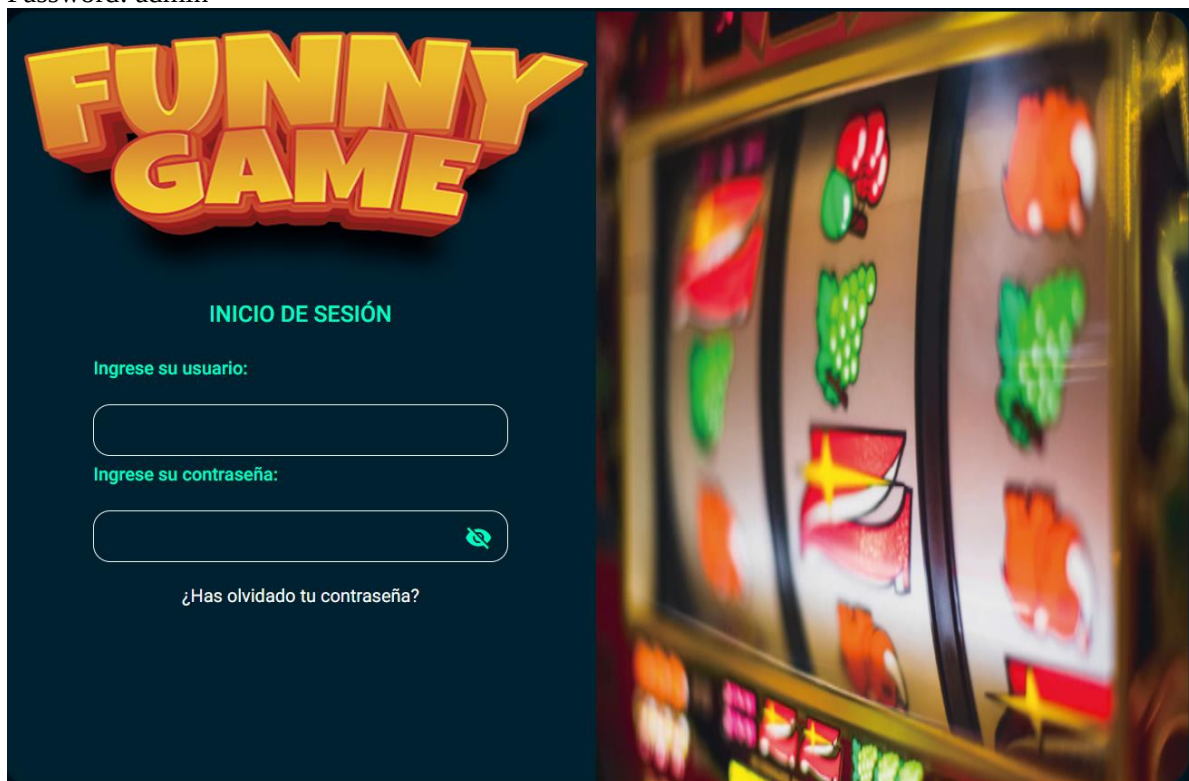


URL's: Production Deployment – Dashboard admin

<https://kioskotouch.pythonanywhere.com/#/login>

User: admin

Password: admin



Continuous Integration with GitHub Actions

Continuous integration (CI) is essential to maintain software quality and stability. In our project, we have implemented GitHub Actions to automate code integration and deployment.

Definition of workflows:

The screenshot displays the GitHub repository interface for '2022-Frontend-Juegos'. The left sidebar shows the file explorer with the '.github/workflows' directory expanded, highlighting the 'ci.yml' file. The main area shows the content of 'ci.yml', which defines a workflow triggered on push. The workflow includes a 'build' job that runs on 'ubuntu-latest' and performs the following steps:

```
10
11 jobs:
12   build:
13     runs-on: ubuntu-latest
14
15     steps:
16     - name: Checkout repository
17       uses: actions/checkout@v2
18
19     - name: Set up Node.js
20       uses: actions/setup-node@v3
21       with:
22         node-version: '14'
23
24     - name: Install dependencies
25       run: npm install
26
27     - name: Build application
28       run: npm run build -- --configuration production
29
30     - name: Run tests
```

Below the code editor, the workflow's execution history is shown. A summary card indicates the workflow was triggered via push 4 days ago by user 'JeffErasLindao' on the 'continuous_integration_dev' branch, resulting in a 'Success' status with a total duration of '2m 36s'. The 'build' job is listed as completed with a duration of '3m'. A 'Workflow file' section shows the 'ci.yml' file with the trigger 'on: push' and a 'build' job status of '2m 26s'.

9. Sprint Review

Public link to the sprint review video:

[https://espol-
my.sharepoint.com/:f:/g/personal/irvmgarz_espol_edu_ec/Eis6DHfz9OpAv938_KudaxUBYWej8Sloj2G-
SDDeYNRFbQ?e=8CyIwe](https://espol.ec/sharepoint.com/:f:/g/personal/irvmgarz_espol_edu_ec/Eis6DHfz9OpAv938_KudaxUBYWej8Sloj2G-SDDeYNRFbQ?e=8CyIwe)

the following activities were carried out:

1. Definition and Initial Development:

- Configuration and loading of games into the system.
- Implementation of the user interface for the client.
- Integration of basic game functionalities such as selection and loading.

2. Review and Adjustments:

- Adjustment of the number of attempts allowed in the games.
- Initial configuration of the probability of winning and loading of prizes.

3. Priorities and Timelines:

- Activities were prioritized with high importance and had an estimated time of 2 to 8 days per task.

User Stories Completed:

1. Client User:

- Play available games.
- Play the game "Roll the Dice".
- Play the game "Guess the Door".
- Play the game "Precision".
- Change games at any time, if no game has been previously started.

10. Acceptance form

https://espolec-my.sharepoint.com/:f:/g/personal/irvmgarz_espolec/Ei59wEW6qPZAu3SX9mWJyvoBhlgPS83a73JpSFW-Xv0lrg?e=zWaits

11. Communications

https://espolec-my.sharepoint.com/:f:/g/personal/irvmgarz_espolec/EsbjXsLSQ-9PvDXZTj9Ss9QBZPteePaJQlkkQPKl48wKrg?e=rpR3cf

12. Installation Guide

https://espolec-my.sharepoint.com/:f:/g/personal/irvmgarz_espolec/EsXQWRjJirJGupcT_lw1WKgB6jtkCM94SB707hxLPfeQ4A?e=Bla2bi

13. Integration Testing

https://espolec-my.sharepoint.com/:f:/g/personal/irvmgarz_espolec/Eltg_eZzG05DqJxbXdMo45EBur0Eg9cpt08QPrTAF0yk7Q?e=ncOcNd